Conditional Expectation

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Overview

Shortcut formula for expectation

Instead of summing over all outcomes, sometimes easier to group outcomes by the blocks defined by the random variable.

$$\mathbb{E}[R] = \sum_{x \in \text{range}(R)} x \cdot \Pr[R = x].$$

Proof:

$\mathbb{E}[R]$	$= \sum_{\omega \in \mathcal{S}} R(\omega) \Pr[\omega]$	defn expt
	$= \sum_{x \in \text{range}(R)} \sum_{\omega \in [R=x]} R(\omega) \Pr[\omega]$	range is partition
	$= \sum_{x \in \text{range}(R)} \sum_{\omega \in [R=x]} x \Pr[\omega]$	defn R, ω
	$= \sum_{x \in \text{range}(R)} x \sum_{\omega \in [R=x]} \Pr[\omega]$	factor out x
	$= \sum_{x \in \text{range}(R)} x \Pr[R = x].$	defn event prob

Median

Not the same as mean or expected value. It's the probability-weighted middle of the set of values.

Definition: The *median* of a random variable R is a value $x \in \text{range}(R)$ such that

$$\Pr[R \leq x] \geq \frac{1}{2}$$

and

$$\Pr[R \geq x] \geq \frac{1}{2}.$$

Definition of conditional expectation

Definition: The conditional expectation of a random variable R given event A is:

$$\mathbb{E}[R|A] := \sum_{r \in \mathrm{range}(R)} r \cdot \Pr[R = r|A].$$

Example: R is the deck of cards example from before: value of card, 1 for ace, 10 for face.

$$\begin{aligned}
&\mathbb{E}[R|R < 10] \\
&= \textstyle\sum_{i=1}^{9} i \Pr[R = i|R < 10] \\
&= \textstyle\sum_{i=1}^{9} i \frac{\Pr[R=i \wedge R<10]}{\Pr[R<10]} \\
&= \textstyle\sum_{i=1}^{9} i \ \frac{1/13}{9/13} \\
&= 1/9 \ \textstyle\sum_{i=1}^{9} i \\
&= 45/9 = 5.
\end{aligned}$$

The height of elephants

How tall are elephants on average? Don't know. Web search didn't help. Conditional expectations can!

$$
\begin{aligned}
\mathbb{E}[H] &= \mathbb{E}[H|\mathrm{Af}] \cdot \Pr[\mathrm{Af}] + \mathbb{E}[H|\mathrm{As}] \cdot \Pr[\mathrm{As}] && \text{split} \\
&= 10 \cdot 0.9 + \mathbb{E}[H|\mathrm{As}] \cdot 0.1 && \text{Google} \\
&= 10 \cdot 0.9 + (\mathbb{E}[H|\mathrm{As} \wedge m] \Pr[m|\mathrm{As}] \\
&\qquad + \mathbb{E}[H|\mathrm{As} \wedge f] \cdot \Pr[f|\mathrm{As}]) \cdot 0.1 && \text{split} \\
&= 10 \cdot 0.9 + (9 \cdot 1/2 + 7 \cdot 1/2) \cdot 0.1 && \text{Google} \\
&= 9.8 && \text{calculator}
\end{aligned}
$$

Memoryless crash time

A program crashes at the end of each hour of use with probability p. Let C be the random variable for the time until a crash occurs. What is $\mathbb{E}[C]$?

Let A be the event that the program crashes after the first hour and $\overline{A}$ be the complementary event. Mean time to failure is

$$\mathbb{E}[C] = \mathbb{E}[C|A] \cdot \Pr[A] + \mathbb{E}[C|\overline{A}] \cdot \Pr[\overline{A}]$$

$\mathbb{E}[C|A] = 1$ because A is the event that the program crashes after one hour.

$\mathbb{E}[C|\overline{A}] = 1 + \mathbb{E}[C]$ because the program runs for an hour, then we're back in the original situation.

$$\begin{aligned}
\mathbb{E}[C] &= 1 \cdot p + (1 + \mathbb{E}[C])(1 - p) \\
\mathbb{E}[C] &= p + 1 + \mathbb{E}[C] - p - p\mathbb{E}[C] \\
p\mathbb{E}[C] &= 1 \\
\mathbb{E}[C] &= 1/p
\end{aligned}$$

Mean time to failure

General principle:

If a system independently fails at each time step with probability p, then the expected number of steps up to the first failure is $1/p$.

More failures

I'm driving down a street with lots of intersections with traffic lights. Each light is (independently) red 50% of the time and green 50% of the time. How many intersections do I expect to drive through before I stop?

failure	red light
hour	intersection
p	$1/2$
$1/p$	2

- Coin flips until heads? 2
- Cards until face card? 13/3 = 4.3 ish
- Kobe (0.85) free throws until basket? 1.18

Expected winnings

D dollars won, w event that you win, $\overline{w}$ event that you lose.

$\mathbb{E}[D] = \mathbb{E}[D|w]\Pr[w] + \mathbb{E}[D|\overline{w}](1 - \Pr[w])$

Game is *fair* if $\mathbb{E}[D] = 0$.

Example: $\mathbb{E}[D|w] = -\mathbb{E}[D|\overline{w}]$, $\Pr[w] = 1/2$.

Example: SAT points (before March 2016). $\mathbb{E}[D|w] = 1$, $\mathbb{E}[D|\overline{w}] = -1/4$, $\Pr[w] = 1/5$.

$1 \cdot .2 + (-1/4) \cdot .8 = 0$.

Bar bet

Three people. Each player puts in \$2. Each player predicts the outcome of a coin flip. Correct guessers split the pot. If all wrong, money returned.

$$
\begin{aligned}
\mathbb{E}[D] \quad = \quad & \mathbb{E}[D|\text{win alone}]\Pr[\text{win alone}] \\
& + \mathbb{E}[D|\text{win with one}]\Pr[\text{win with one}] \\
& + \mathbb{E}[D|\text{win with both}]\Pr[\text{win with both}] \\
& + \mathbb{E}[D|\text{lose}]\Pr[\text{lose}] \\
& + \mathbb{E}[D|\text{all lose}]\Pr[\text{all lose}]
\end{aligned}
$$

$$
\begin{aligned}
\mathbb{E}[D] \quad = \quad & 4 \cdot \Pr[\text{win alone}] \\
& + 1 \cdot \Pr[\text{win with one}] \\
& + 0 \cdot \Pr[\text{win with both}] \\
& - 2 \cdot \Pr[\text{lose}] \\
& + 0 \cdot \Pr[\text{all lose}]
\end{aligned}
$$

Random guesses

$$\begin{aligned}\mathbb{E}[D] = \; & 4 \cdot \Pr[\text{win alone}] \\ & +1 \cdot \Pr[\text{win with one}] \\ & +0 \cdot \Pr[\text{win with both}] \\ & -2 \cdot \Pr[\text{lose}] \\ & +0 \cdot \Pr[\text{all lose}]\end{aligned}$$

Everyone guesses randomly.

- $\Pr[\text{win alone}] = 1/8$
- $\Pr[\text{win with one}] = 1/4$
- $\Pr[\text{win with both}] = 1/8$
- $\Pr[\text{lose}] = 3/8$
- $\Pr[\text{all lose}] = 1/8$

$4 \cdot 1/8 + 1 \cdot 1/4 + 0 \cdot 1/8 - 2 \cdot 3/8 + 0 \cdot 1/8 = 0$. Fair!

Collusion

$$\begin{aligned}\mathbb{E}[D] \quad = \quad & 4 \cdot \Pr[\text{win alone}] \\ & +1 \cdot \Pr[\text{win with one}] \\ & +0 \cdot \Pr[\text{win with both}] \\ & -2 \cdot \Pr[\text{lose}] \\ & +0 \cdot \Pr[\text{all lose}]\end{aligned}$$

The other two collude to make opposite guesses.

- $\Pr[\text{win alone}] = 0$
- $\Pr[\text{win with one}] = 1/2$
- $\Pr[\text{win with both}] = 0$
- $\Pr[\text{lose}] = 1/2$
- $\Pr[\text{all lose}] = 0$

$4 \cdot 0 + 1 \cdot 1/2 + 0 \cdot 0 - 2 \cdot 1/2 + 0 \cdot 0 = -0.5$. I lose!

Irrational bookies

Three horses, a, b, c. For any horse you can bet that it will win or lose. I give you odds:

- 1:1 odds for a
- 3:1 odds against b
- 4:1 odds against c

Should you bet with me?

Yes—I'm irrational! You should bet: \$100 on a, \$50 on b, \$40 on c. You spend \$190. No matter which horse wins, you win \$200.

Be careful with logical combinations of events, too. More in a couple weeks.